Kew

Final report | Team 254 | Megathon X: Singularity
Lloyds Problem Statement 2 | 11 October 2026

Solution

Kew helps homeowners understand their home's energy performance and compare practical retrofit packages within a budget. It also gives Lloyds-oriented advisors a portfolio view of properties and improvement potential. The result is a working, explainable prototype that connects an interactive web experience with deterministic calculations and a grounded AI advisor.

The homeowner journey runs from property selection and assessment to retrofit planning and before-and-after comparison. Instead of presenting a single opaque recommendation, Kew shows the evidence, assumptions, suitability and tradeoffs behind each option.

Core features

Home assessment. Editable property details, annual fuel use, data provenance and explicit missing-data warnings. A Green Home Score from 0 to 100 includes a classification and seven weighted component contributions.

Retrofit planning. Loft insulation, double glazing, heating controls, heat pumps and solar PV are checked for feasibility. Budget-aware packages are simulated sequentially, then ranked for energy reduction, carbon reduction or shortest payback.

Scenario comparison. Current and proposed homes are compared on purchased electricity and gas, net energy, carbon, installation cost, annual bill savings, payback and supported indicative value ranges.

Advisor and portfolio. Ask Kew explains recorded calculations and routes what-if questions to fresh backend simulations. Portfolio views use the same assessment outputs to compare homes and improvement opportunities.

User experience. A responsive Next.js workspace includes an opening autumn leaf sequence, an interactive 3D house introduction, clear navigation, comparison tables and a contact page.

Prototype scope: supplied and generated synthetic data. Outputs support comparison and explanation; they do not establish measured savings or observed-market property values.

Approach

Transparent calculations first. AI explains the evidence.

System design

The Next.js frontend sends property and scenario requests through its backend proxy to a FastAPI service. Python modules handle assessment, feasibility, energy simulation, carbon, package optimisation and valuation. SQLite preserves property records and associated physical inputs, costs and tariffs.

Flow: property inputs → validation → baseline assessment → feasibility checks → sequential retrofit simulation → package optimisation → comparison and grounded explanation.

Deterministic assessment and simulation

The score combines EPC (25%), purchased-energy intensity (20%), insulation (15%), heating (15%), windows (10%), renewable contribution (10%) and controls (5%). Missing required inputs make the overall score unavailable while known components remain visible. Weights are not redistributed.

The simulator carries physical state through each upgrade so interactions are reflected in the final package. It retains hot-water and appliance demand, records additional heat-pump electricity, and applies solar to purchased imports. Carbon is calculated separately for electricity and gas using documented 2026 DESNZ factors. Bills require explicit tariffs; nonpositive savings do not produce a misleading payback.

With five measures, the optimiser can consider all 32 subsets. Accepted packages must satisfy the budget and confirmed feasibility. Conditional and infeasible measures remain distinguishable, and alternative objective winners expose cost, energy and carbon tradeoffs.

AI engineering

The advisor uses a configurable provider layer with hosted and local Ollama support and a deterministic fallback. Calculations stay in backend tools. What-if requests invoke feasibility, simulation or optimisation rather than asking the language model to invent results. Responses are grounded in structured outputs, with checks for unsupported numeric claims and clear handling of partial or unavailable evidence.

Valuation and uncertainty

Snapshot valuation and synthetic scenario impact are treated separately. The impact model uses before-and-after purchased-energy intensity while preserving the recorded EPC. Ridge and gradient-boosting models are compared using separate fit, selection, calibration and test groups. Unsupported inputs return structured unavailable results. Synthetic value changes are labelled illustrative, with separately calibrated paired uncertainty.

Results

Recorded repository evidence, with limits stated explicitly.

The figures below come from the saved reproduction and evaluation reports. They are recorded results, not a new full test run performed while preparing this PDF.

Check / measure	Recorded result
Software verification	224 Python tests + 34 subtests; 13 frontend tests
Reproduction pipeline	All 11 commands passed, including typecheck and build
Held-out retrofit evaluation	24 synthetic homes; 119 fuel cases; 120 feasibility decisions
Rule-reference agreement	Feasibility accuracy 100%; fuel/carbon MAE below 1e-7
Package selection	Ranking and optimiser each matched 72/72 reference cases
Synthetic price prediction	MAE £23,642; RMSE £32,866; R^2 0.992
Price model comparison	Ridge MAE £67,422; median baseline MAE £284,196
Synthetic paired value change	MAE £5,655; interval coverage 86.4% at nominal 90%

What these results demonstrate

The prototype produces reproducible, internally consistent assessments and package decisions under its documented assumptions. Separate property groups prevent fit/test overlap in the 1,800-pair synthetic impact dataset. Its 360 untouched test properties achieved 90.8% coverage for nominal 90% price bands. The paired change interval undercovers at 86.4%, so uncertainty calibration remains an improvement area.

Limits and next steps

Synthetic price labels and rule-reference agreement do not prove real-world retrofit savings, causal value uplift or observed-market accuracy. Some after-scores remain unavailable without a supported renewable-category mapping. Historical Ollama evidence includes 25 responses with no unsupported numeric claims in its numeric audit, but predates the latest impact feature; human explanation-quality review remains pending.

Next steps are independent measured-energy and observed-sales validation, stronger interval calibration, seasonal heat-pump and solar modelling, and completion of the human advisor review.

Evidence in project/: docs/improvement_report.md; docs/scoring_methodology.md; evaluation/results/current_pipeline/submission_verification.json; evaluation/results/held_out_retrofits/metrics.json; ml/reports/synthetic_impact/valuation_metrics.json.